

# Xi GUO

— im.guoxi@gmail.com — +1 (406)402-2751 — zerocath.github.io/ — www.linkedin.com/in/xigu0815

## SUMMARY

Software Engineer with 2+ years of industry experience building ML infrastructure for search and recommendation at one of China's largest e-commerce platforms. Experienced in large-scale distributed training, ML frameworks, and systems performance optimization. Currently pursuing an M.S. in ECE at the University of Washington (graduating June 2028).

## EDUCATION

### University of Washington

*Master of Science* in Electrical and Computer Engineering

Seattle, WA

Sep 2026 — Jun 2028

### Huazhong University of Science and Technology

*Bachelor of Engineering* in Computer Science and Technology

Wuhan, China

Sep 2019 — Jun 2023

## SKILLS

**Languages & Frameworks:** C/C++, Python, Go, Java, Bash, PyTorch, TensorFlow, TorchRec, CUDA, Triton, DDP, FSDP

**Systems & Tools:** Parameter Servers, Zookeeper, Docker, HDFS, Linux (Kernel/Profiling), NCCL, Kubernetes

## PROFESSIONAL EXPERIENCE

### PDD Holdings (Pinduoduo & Temu)

*Software Engineer, Machine Learning Infrastructure*

Shanghai, China

Jul 2023 — Nov 2025

#### *Large-scale training for generative recommendation*

- Built **PyTorch/TorchRec** training workflows for a generative recommender (HSTU) serving **300M+ DAU**, scaling billion-parameter sparse training across recall, CTR, and CVR tasks with DDP and TorchRec DMP.
- Contributed to a **pipelined** training framework overlapping CPU data preprocessing, sparse embedding lookup, and forward/backward computation, and parameter updates, improving training throughput by **40%+**.
- Optimized sparse embedding communication by deduplicating feature IDs before TorchRec embedding lookup, improving 32 H100 GPU end-to-end throughput by **25%**.

#### *Training Framework Migration (TensorFlow → PyTorch)*

- Migrated a recommendation model training pipeline from TensorFlow to PyTorch on **multi-node H100 clusters**, re-implementing data loading, sharded checkpointing, and cross-framework consistency checks.
- Integrated custom **C++ sparse Parameter Server** as a CPU backend to offload massive read-only embedding tables, avoiding GPU memory constraints for memory-intensive downstream models.
- Optimized **data prefetching** and CPU-GPU coordination, increasing GPU utilization by **30%+** and reducing training time by **15%** versus the legacy TensorFlow pipeline.

#### *Semi-Synchronous Training Mechanism*

- Extended TensorFlow SyncReplicaOptimizer with a semi-sync mechanism, implementing global batch aggregation, token-based **gradient admission** and staleness control while allowing workers to progress independently.
- Improved training throughput by **60%+** over synchronous training on a production CVR model, reaching **85%** of asynchronous training throughput while observing a **0.3%** AUC improvement.
- Supported parameter-level update strategies by TensorFlow graph modifications to meet model-specific requirements.

#### *Distributed Training Infrastructure*

- Built multi-node training startup coordination with **ZooKeeper** and RPC, using CAS-based leader election to assign a primary worker for code distribution and cluster metadata collection.
- Coordinated worker readiness and generated TF\_CONFIG for multi-node training jobs, enabling reliable cluster bring-up.
- Developed **backend APIs**(Python/Java) for training job submission, resource management, and access control, and implemented proxy routing to unify service endpoints.

### Sensetime

*Software Engineer Intern*

Shanghai, China

Jul 2022 — Sep 2022

- Developed Golang (Gin) backend services and **REST APIs** for an enterprise deep learning platform, supporting algorithm management, training workflows, and access control.
- Developed and maintained Helm-based deployment workflows for **Kubernetes** services and training jobs, including chart validation, environment configuration, and release deployment.
- Integrated pretrained models and training pipelines into the platform, implementing model configuration validation and supporting model artifact management with object storage.